

UNIT-IV

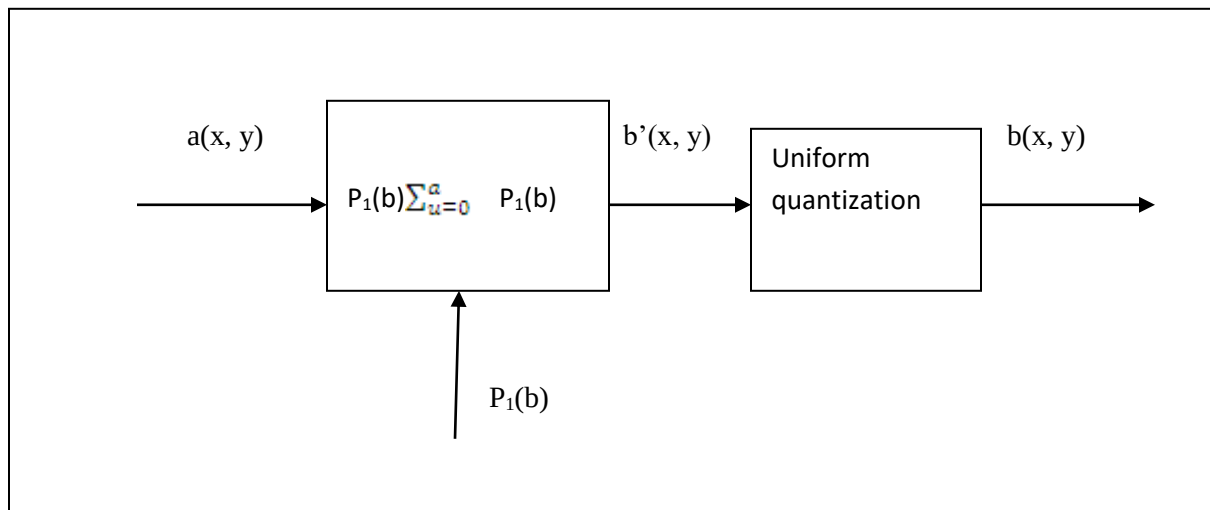
Image Enhancement and Restoration: Histogram Modification Techniques – Image Smoothinging – Image Sharpening – Image Restoration – Degradation Model – Noise Models – Spatial Filtering – Frequency Domain Filtering.

HISTOGRAM MODELING:

It is one of the powerful techniques in image enhancement. The histogram of an image represents the relative frequency of occurrence of the various gray levels in the image. Histogram may be modified to have a desired shape. This is useful in stretching the low-contrast levels of images with narrow histograms.

HISTOGRAM EQUALIZATION:

The histogram equalization operation is shown in the figure. A cumulative histogram of the function is found and then uniform quantization is carried out.



Histogram equalization transformation

This technique for equalizing the histogram gives the best possible dynamic range and strong contrast. An image is more visible if we use this technique.

Let the variable r represent the gray levels in the image to be enhanced. Initially assume that pixel values are continuous quantities that have been normalized. So they lie in the interval $[0,1]$ with $r=0$ represent blank and $r=1$ represent white.

For any value r in the interval $[0,1]$ the transformation is given by

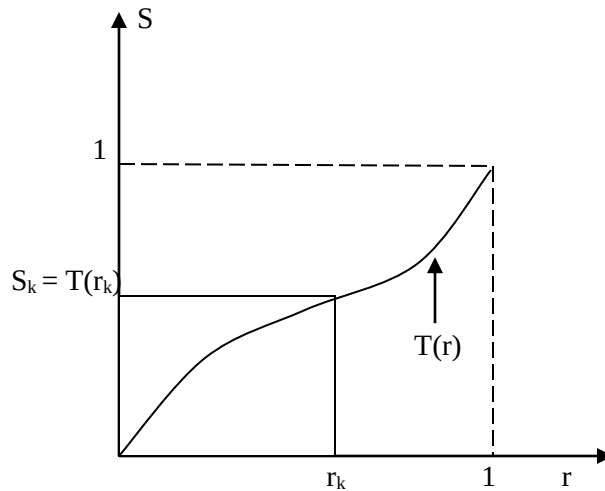
$$S=T(r)$$

Which produce a level for every pixel value r in the original image.

This equation satisfied the **conditions**.

- (i) $T(r)$ is single valued and monotonically increased in the interval $0 \leq r \leq 1$.
(i.e. preserves the order from blank to white in the gray scale)
- (ii) $0 \leq T(r) \leq 1$ for $0 \leq r \leq 1$
(i.e. guarantees a mapping that is consistent with the allowed range of pixel values)

FIGURE: Shows the transformation function satisfying these conditions.



A gray level transformation function. The inverse transformation is given by

$$r = T^{-1}(S) \quad 0 \leq S \leq 1$$

The gray levels in an image may be as random quantities in the interval $[0,1]$. If they are continuous variables, the original and transformed gray levels can be characterized by their probability density functions $P_r(r)$ and $P_s(s)$ respectively.

If $P_r(r)$ and $T(r)$ are known and $T^{-1}(S)$ satisfies the conditions (i), the probability density function of the transformed gray level is

$$P_s(s) = [P_r(r) dr/ds]_{r=T^{-1}(s)}$$

Consider the transformation

$$S = T(r) = \int_0^r P_r(w) dw \quad 0 \leq r \leq 1$$

Where w is the dummy variable of integration.

The right side value of equation represents the cumulative distribution function (CDF) if r . Conditions (i) and (ii) are satisfied by this transformed function because the CDF increases monotonically from 0 to 1 as a function of r .

From equation ,the derivative of S with respect to r is

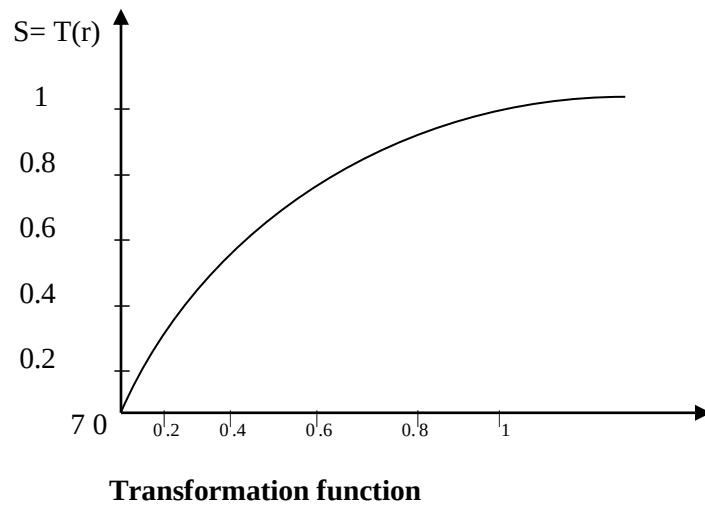
$$ds/dr = P_r(r)$$

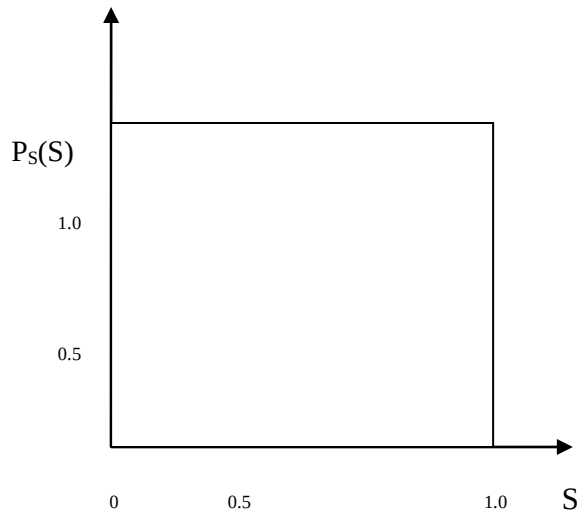
substitute dr/ds value into equation

$$\begin{aligned} P_s(s) &= [(P_r(r) \cdot 1) / P_r(r)]_{r=T^{-1}(s)} \\ &= [1]_{r=T^{-1}(s)} \\ &= 1, 0 \leq s \leq 1 \end{aligned}$$

Which is a uniform density function.

FOR EXAMPLE:





Resulting uniform density

From the fig the levels r have the probability density function $P_r(r)$ which is given by

$$P_r(r) = \begin{cases} -2r + 2 & 0 \leq r \leq 1 \\ 0 & \text{Otherwise} \end{cases}$$

Substitute this value in equation which gives the transformation function.

$$\begin{aligned} S = T(r) &= \int_0^r (-2w + 2) dw \\ &= -r^2 + 2r \end{aligned}$$

Inverse transform is

$$\begin{aligned} r &= T^{-1}(S) \\ &= 1 \pm \sqrt{1 - S} \end{aligned}$$

Since r lies in the interval $[0, 1]$

$$r = T^{-1}(S) = 1 - \sqrt{1 - S} \text{ is valid.}$$

The probability density function is given by

$$P_s(S) = \left(P_r(r) \frac{dr}{ds} \right)_{r = T^{-1}(S)}$$

$$= \left(-2r + 2 \right) \frac{dr}{ds} \quad r = 1 - \sqrt{1-s}$$

$$= [(-2[1-\sqrt{1-s}]+2)d(1-\sqrt{1-s})]/ds]$$

$$= [-2+2\sqrt{1-s}+2].d(1-\sqrt{1-s})/ds$$

$$= 2\sqrt{1-s}.d(1-\sqrt{1-s})/ds$$

$$=1 \quad 0 \leq s \leq 1$$

which is a uniform density in the desired range.

For gray levels that take on discrete values, then p.d.f. is

$$P_r(r_k) = n_k/n \quad 0 \leq r_k \leq 1 \text{ and } K=0,1,2,\dots,L-1$$

Where, L is the number of levels

$P_r(r_k)$ is the probability of the Kth gray level.

n_k is the number of times this level appear in the image.

n is the total no. of pixels in the image.

A plot of $P_r(r_k)$ Vs r_k is called 'histogram' and the technique used for obtaining a uniform histogram is known as histogram equalization or histogram linearization.

The discrete form of equation (2) is

$$S_k = T(r_k) = \sum_{j=0}^k n_j/n$$

$$= \sum_{j=0}^k P_r(r_j) \quad 0 \leq r_k \leq 1 \text{ and } K=0,1,2,\dots,L-1$$

The inverse transformation is

$$r_k = T^{-1}(S_k) \quad 0 \leq S_k \leq 1$$

HISTOGRAM SPECIFICATION :

The histogram equalization does not used in image enhancement applications because it is capable of generating only one result i.e. uniform histogram.

Sometimes the ability to specify particular histogram shapes capable of highlighting certain gray level ranges in an image is desirable.

Let $P_r(r)$ and $p_d(d)$ be the original and desired p. d .f. respectively.

The histogram equalization of original image is

$$S = T(r) = \int_0^r P_r(w) dw$$

The histogram equalization of desired image is

$$V = G(d) = \int_0^d P_d(w) dw$$

The inverse process is $d = G^{-1}(V)$

which gives the d of the desired image.

$P_s(S)$ and $P_v(V)$ are the identical uniform densities, because, the final result is independent of the density inside the integral.

The procedure is

- (i) To equalize the levels of the original image using transformation equation i.e. (2)
- (ii) Specify the desired density function and get the transformation function $G(d)$ using equation (2.3.3).
- (iii) Apply the inverse transformation function, $d=G^{-1}(S)$ to the level got in step (i).

This procedure gives a processed value of original image with the new gray levels characterized by the specified density $P_d(d)$.

IMAGE SMOOTHING

Smoothing may be distinguished from the related and partially overlapping concept of curve fitting in the following ways:

Curve fitting often involves the use of an explicit function form for the result, whereas the immediate results from smoothing are the "smoothed" values with no later use made of a functional form if there is one;

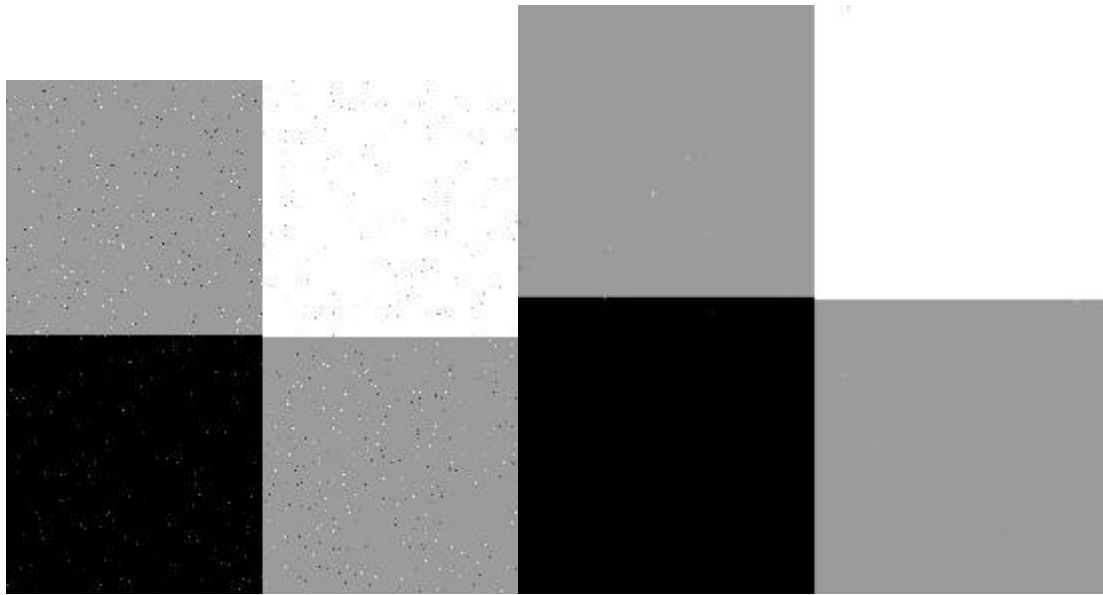
The aim of smoothing is to give a general idea of relatively slow changes of value with little attention paid to the close matching of data values, while curve fitting concentrates on achieving as close a match as possible.

Smoothing methods often have an associated tuning parameter which is used to control the extent of smoothing.

Median Filtering

Median filtering is needed to remove impulse noise from an image. A pixel intensity is replaced with the median of pixel intensities within a window centered at that pixel. If a part of the window falls outside the image, intensities within the portion of the window inside the image is used.

Circular windows are used to make smoothing independent of the image orientation. An example of median filtering by this software is given below.



(a)

(b)

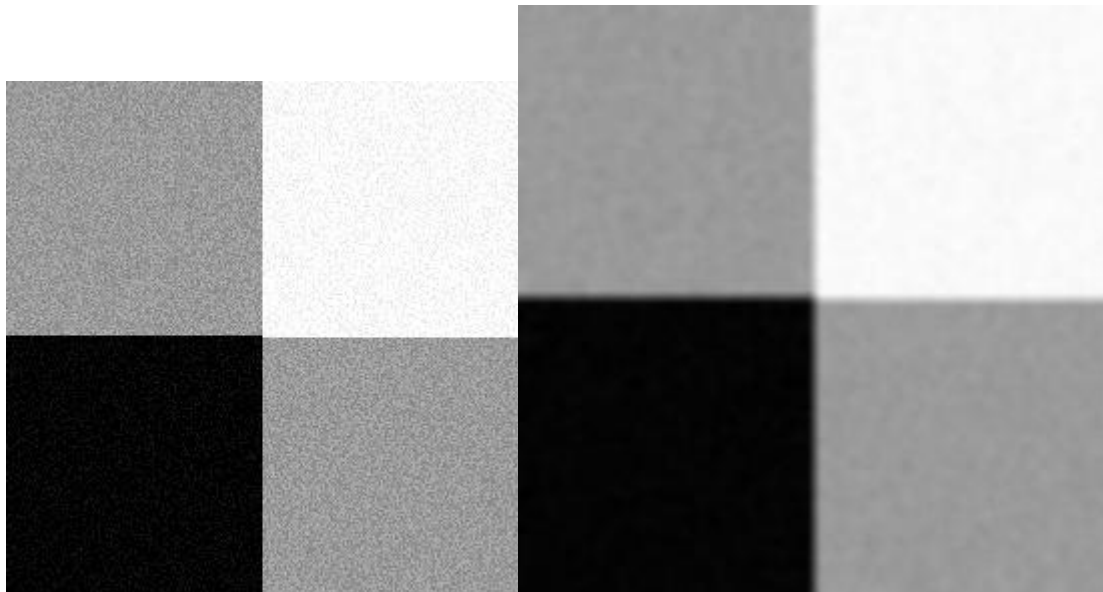
Fig. 1. (a) An image containing impulse noise.

(b) Smoothing the image by median filtering using a filter of radius 1 pixel.

Mean Filtering

Mean filtering is image averaging and the value at a pixel in the output is equal to the average of values within a circular window centered at the pixel in the input.

The window size determines the neighbourhood size where averaging is performed. As the window size is increased, more noise is removed, but that at the same time blurs the image more. An example of mean filtering by this method is given below.



(a)

(b)

Fig. 2. (a) An image containing zero-mean noise.
(b) Image obtained by mean filtering using a filter of radius 4 pixels.

Gaussian Filtering

A Gaussian is an ideal filter in the sense that it reduces the magnitude of high spatial frequencies in an image proportional to their frequencies. That is, it reduces magnitude of higher frequencies more. This will be at the cost of more computation time when compared to mean filtering.

A Gaussian extends to infinity in all directions, but because it approaches zero exponentially, it can be truncated three or four standard deviation away from its center without affecting the result noticeably. Speed up is achieved by splitting a 2-D Gaussian into two 1-D Gaussians, $G(x,y) = G(x) G(y)$, and carrying out filtering in 1-D, first row by row and then column by column. An example of Gaussian filtering is given below.

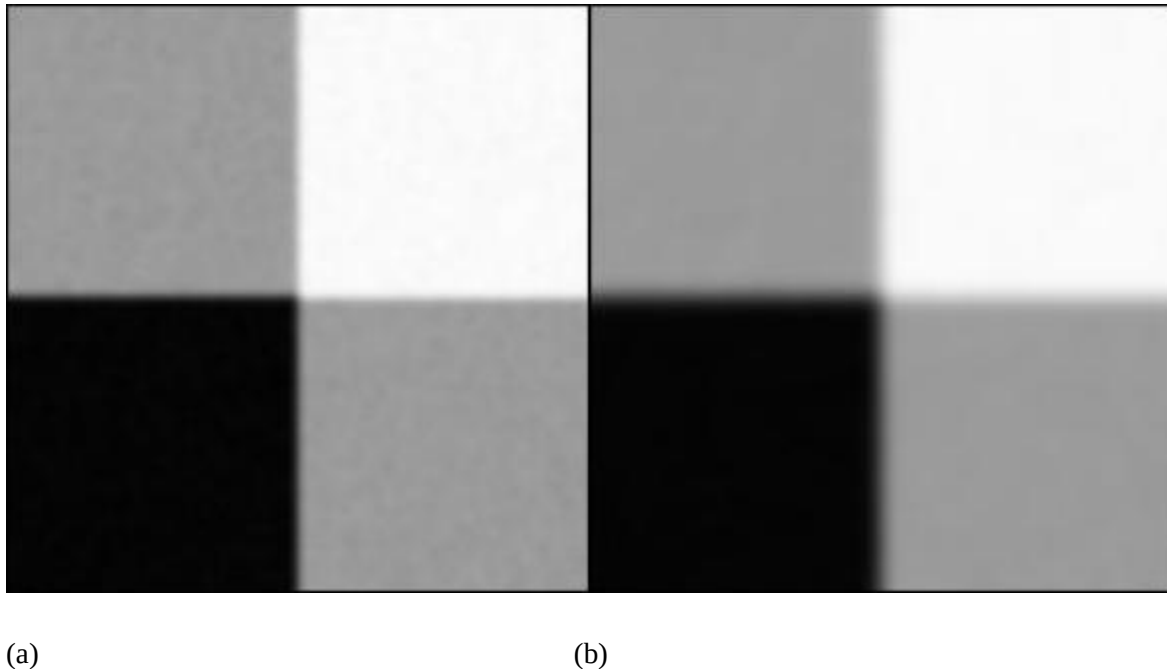


Fig. 3. (a) Image that containing noise.
(b) Smoothing the image shown in Fig. 2a by Gaussian filters of standard deviations 2 and 4 pixels, respectively.

Conservative Smoothing

Brief Description

Conservative smoothing is a noise reduction technique that derives its name from the fact that it employs a simple, fast filtering algorithm that sacrifices noise suppression power in order to preserve the high spatial frequency detail (e.g. sharp edges) in an image. It is explicitly designed to remove noise spikes --- i.e. isolated pixels of exceptionally low or high pixel intensity (e.g. salt and pepper noise) and is, therefore, less effective at removing additive noise (e.g. Gaussian noise) from an image.

How It Works?

Like most noise filters, conservative smoothing operates on the assumption that noise has a high spatial frequency and, therefore, can be attenuated by a local operation which makes each pixel's intensity roughly consistent with those of its nearest neighbors.

However, whereas mean filtering accomplishes this by averaging local intensities and median filtering by a non-linear rank selection technique, conservative smoothing simply ensures that each pixel's intensity is bounded within the range of intensities defined by its neighbors.

This is accomplished by a procedure which first finds the minimum and maximum intensity values of all the pixels within a windowed region around the pixel in question. If the intensity of the central pixel lies within the intensity range spread of its neighbors, it is passed on to the output image unchanged.

However, if the central pixel intensity is greater than the maximum value, it is set equal to the maximum value; if the central pixel intensity is less than the minimum value, it is set equal to the minimum value.

Figure 1 illustrates this idea.

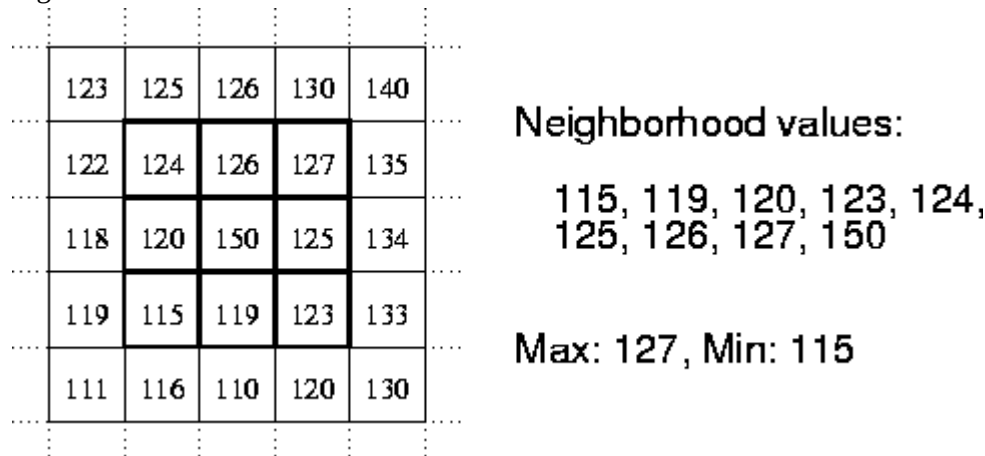


Figure 1 Conservatively smoothing a local pixel neighborhood. The central pixel of this figure contains an intensity spike (intensity value 150). In this case, conservative smoothing replaces it with the maximum intensity value (127) selected amongst those of its 8 nearest neighbors.

If we compare the result of conservative smoothing on the image segment of Figure 1 with the result obtained by mean filtering and median filtering, we see that it produces a more subtle effect than both the former (whose central pixel value would become 125) and the latter (124). Furthermore, conservative smoothing is less corrupting at image edges than either of these noise suppression filters.

IMAGE SHARPENING

Sharpening is one of the most impressive transformations you can apply to an image since it seems to bring out image detail that was not there before.

What it actually does, however, is to emphasize edges in the image and make them easier for the eye to pick out -- while the visual effect is to make the image seem sharper, no new details are actually created. Paradoxically, the first step in sharpening an image is to blur it slightly. Next, the original image and the blurred version are compared one pixel at a time.

If a pixel is brighter than the blurred version it is lightened further; if a pixel is darker than the blurred version, it is darkened. The result is to increase the contrast between each pixel and its neighbors.

The nature of the sharpening is influenced by the blurring radius used and the extent to which the differences between each pixel and its neighbor are exaggerated.

Sharpening Methods

Picture Window offers three sharpening techniques: Sharpen, Heavy Sharpen, and Unsharp Mask. Sharpen and Heavy Sharpen are the most straightforward methods.

They are computed similarly, but Sharpen blurs the pixels in the image based on only its four closest neighbors and exaggerates the differences moderately while heavy Sharpen blurs by using the eight nearest neighbors and exaggerates the differences more. In either case, you can moderate the effect by setting the Amount slider back from its 100% maximum value.

The effects of Sharpen and Heavy Sharpen are illustrated below:



Sharpen

Heavy Sharpen

Oversharpening

If you continue to sharpen an image, several things happen:

1. Edges become unnaturally pronounced—dark objects are outlined with light halos and light objects are outlined with dark halos.
2. Normally invisible noise in the image is amplified and starts to show up as texture in areas that looked smooth in the original images. This can create an undesirable graininess in parts of photographs that should be smooth like clouds and skies.
3. Extreme sharpening causes the image to break up as each individual pixel stands out more and more from its neighbors.

Thus as you progressively sharpen a slightly blurry image, it first starts to look better, but then as you begin to oversharpen it, it starts to look worse again. Once you know what oversharpening looks like, you will be able to spot it on the screen and back off a little.



Heavy Sharpen

Twice Heavy Sharpen Three Times

Using the Sharpen transformation

The best way to set up Picture Window's Sharpen Transformation is to zoom the input image in to a magnification factor of 1:1 or even 2:1 and then zoom the Preview window in to the same magnification factor and scroll the two windows as necessary to display the same part of the image.

Resize and reposition the two windows so they are next to each other across the top of the screen. This lets you see every detail at the individual pixel level in both the original image and the sharpened version.

As you preview the results of various settings, you will then be able to compare the original and sharpened versions easily. If necessary, you can scroll both windows to see the effect of the transformation on other parts of the image.

For slightly blurred images, the Sharpen or Heavy Sharpen options often work well.

If the effect is too strong, you can always move the Amount slider to the left to reduce the amount of sharpening. If you want to sharpen just part of an image, leaving the rest alone, you will need to create a mask that isolates the region of interest and then select this mask into the Amount control. Making and using masks is covered in the Picture Window manual and help file.

Unsharp Masking

Unsharp masking is the most powerful sharpening method Picture Window supports, however it is a little more complicated to use. When you select Unsharp Masking, the Sharpen dialog box expands to add two additional sliders for Radius and Threshold.

The Radius slider lets you control the amount of blurring. Generally you should set the radius to correspond to the degree to which the original image is blurred. The blurrier the image, the higher the radius you need to select. Choosing too large a radius creates a sort of ghosting effect around the edges of objects; if the radius is too small, the sharpening effect is minimized.

The Threshold setting lets you restrict the sharpening action to only those pixels whose difference from their neighbors exceeds a specified threshold value.

The idea behind setting the threshold value is to select a value that still brings out edge detail without creating unwanted texture in smooth areas like clouds or clear blue skies. In the image detail below, you

can see how Unsharp Mask with a threshold of zero sharpens the tree silhouette, but also brings out the film grain and scanning noise in the sky area.

Increasing the threshold to 20 leaves the sky mostly untouched but still makes the tree stand out against its background.



Unsharp Mask
Threshold 0

Unsharp Mask
Threshold 20

Note: this example has been deliberately oversharpened to emphasize the effects of the threshold setting.

IMAGE RESTORATION

Image restoration is similar to the process of image enhancement. Image restoration is the process that reconstructs an image that has been degraded by using some prior knowledge of degradation phenomenon.

Restoration is the process that restores the image. The restoration techniques use modeling the degradation and applying the inverse process to recover the original image.

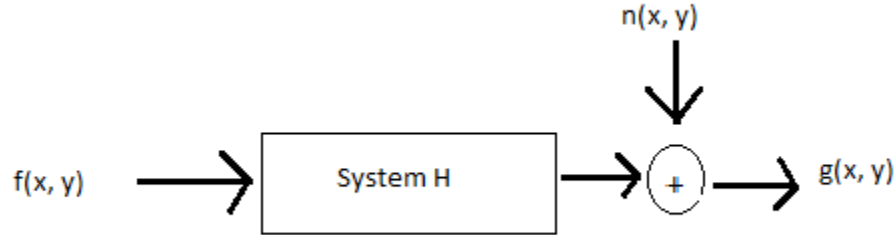
Digital images can be degraded in many ways,

For example,

- i) In analog to digital converter (ADC), linear noise may arise.
- ii) Computer can cause random point degradations.
- iii) Interference from radio source, electric motor of such frequency will produce lines at regular intervals on the screen. If the object moves, as it is being exposed, the image will be blurred.

The algorithms for detecting and removing degradation in the spatial domain are convolution, modeling, matrix and re-sampling techniques. In frequency domain, Fourier reconstruction algorithm is used.

Model of Image Degradation



The input-image $f(x, y)$ is applied to the system 'H'. The additive noise $n(x, y)$ is also added to the system output to produce a degraded image $g(x, y)$.

The input-output relationship is expressed as

$$g(x, y) = H[f(x, y)] + n(x, y) \quad \rightarrow(1)$$

Properties of degradation model:

i)Linearity:

Assume $n(x, y) = 0$ in eqn 1

$$\therefore g(x, y) = H[f(x, y)] \quad \rightarrow(2)$$

H is linear if

$$H[k_1 f_1(x, y) + k_2 f_2(x, y)] = k_1 H[f_1(x, y)] + k_2 H[f_2(x, y)] \quad \rightarrow(3)$$

Where k_1 and k_2 are constants, and $f_1(x, y)$ and $f_2(x, y)$ are any two input images. The linear operator possesses both additive and homogeneity property.

ii)Additive:

If $k_1=k_2=1$ in eqn 3. It is called additive property.

$$H[f_1(x, y) + f_2(x, y)] = H[f_1(x, y)] + H[f_2(x, y)] \quad \rightarrow(4)$$

It states that, if H is a linear operator, the response to a sum of two inputs is equal to the sum of the two responses.

iii)Homogeneity:

It states that the response to a constant multiple of any input is equal to the response to that input multiplied by the same constant.

i.e., $f_2(x, y)=0$ in eqn (3)

$$H[k_1 f_1(x, y)] = k_1 H[f_1(x, y)] \rightarrow (5)$$

Which is called the homogeneity property

iv) Position invariant (or) Space invariant:

An operation having the input-output relationship $g(x, y) = H[f(x, y)]$ is said to be “position invariant” if

Sub $x = x - \alpha, y = y - \beta$

$$H[f(x - \alpha, y - \beta)] = g(x - \alpha, y - \beta) \rightarrow (6)$$

For any $f(x, y)$ and any α and β .

Position invariant indicates that, the response at any point in the image depends only on the value of the input at that point and not on the position of the point.

Degradation model for continuous function:

When the image $f(x, y)$ is convolved with the two dimensional impulse function $\delta(x, y)$, then it can be represented by

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \delta(x - x', y - y') dx' dy' \rightarrow (7)$$

We use the dummy variable α and β and the above equation can be written as

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) \delta(x - \alpha, y - \beta) d\alpha d\beta \rightarrow (8)$$

sub equation (8) in (1)

$$g(x, y) = H[f(x, y)] + n(x, y)$$

$$f(x, y) = H \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) \delta(x - \alpha, y - \beta) d\alpha d\beta \right] + n(x, y) \rightarrow (9)$$

If $n(x, y) = 0$ and if H is a linear operator & we extend the additivity property to the integrals, then

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} H[f(\alpha, \beta) \delta(x - \alpha, y - \beta)] d\alpha d\beta \rightarrow (10)$$

Since $f(\alpha, \beta)$ is independent of x and y and from the homogeneity property,

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) H[\delta(x - \alpha, y - \beta)] d\alpha d\beta \quad \rightarrow (11)$$

We know the “impulse response” H can be represented as

$$h(x, \alpha, y, \beta) = H[\delta(x - \alpha, y - \beta)] \quad \rightarrow (12)$$

Then g(x, y) can be represented by

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x, \alpha, y, \beta) d\alpha d\beta \quad \rightarrow (13)$$

This equation is called as “superposition (or Fredholm) integral of the first kind”.

If H is a position invariant, then,

$$H[\delta(x - \alpha, y - \beta)] = h(x - \alpha, y - \beta) \quad \rightarrow (14)$$

Then equation (13) becomes,

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x - \alpha, y - \beta) d\alpha d\beta \quad \rightarrow (15)$$

In the presence of additive noise, the expression describing a linear degradation model becomes

$$g(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\alpha, \beta) h(x, \alpha, y, \beta) d\alpha d\beta + n(x, y) \quad \rightarrow (16)$$

This is the equation for degradation model for continuous image function f(x, y).

Discrete degradation model:

Digital images are used in most computer based applications and therefore it is needed to formulate discrete degradation model.

Let two functions f(x) and h(x) are sampled uniformly to form arrays of dimensions A and B respectively.

The discrete variable ranges 1, 1, 2.... A-1 for f(x) and 0, 1, 2... B-1 for h(x).

To avoid overlapping in the individual periods, a common period “M ≥ A + B - 1” is selected and in the individual functions zeros are appended after the interval A and B in the functions f(x) and h(x) and they are represented as f_e(x) and h_e(x) i.e., extended functions.

The convolution of f_e(x) and h_e(x) is given by

$$g_e(x) = \sum_{m=0}^{M-1} f_e(m) h_e(x-m) \quad \rightarrow (17)$$

$$\text{for } x = 0, 1, 2, \dots, M-1$$

Both f_e(x) and h_e(x) are assumed to have a period equal to M. g_e(x) also has this period. Then, the above equation is expressed in the matrix notation as

$$g = Hf \quad \rightarrow (18)$$

where f and g are M dimensional column vectors.

$$f = \begin{pmatrix} f_e(0) \\ f_e(1) \\ \vdots \\ f_e(M-1) \end{pmatrix} \quad \rightarrow (19)$$

and

$$g = \begin{pmatrix} g_e(0) \\ g_e(1) \\ \vdots \\ g_e(M-1) \end{pmatrix} \quad \rightarrow (20)$$

and H is the M x M matrix

$$H = \begin{pmatrix} h_e(0) & h_e(-1) & h_e(-2) & \dots & h_e(-M+1) \\ h_e(1) & h_e(0) & h_e(-1) & \dots & h_e(-M+2) \\ h_e(2) & h_e(1) & h_e(0) & \dots & h_e(-M+3) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ h_e(M-1) & h_e(M-2) & h_e(M-3) & \dots & h_e(0) \end{pmatrix} \quad \rightarrow (21)$$

Assume the function $h_e(x)$ has the priority property, then $h_e(x) = h_e(M + x)$. Then the matrix “H” is written as

$$H = \begin{pmatrix} h_e(0) & h_e(M-1) & h_e(M-2) \dots & h_e(1) \\ h_e(1) & h_e(0) & h_e(M-1) \dots & h_e(2) \\ \dots & \dots & \dots & \dots & \dots \\ h_e(M-1) & h_e(M-2) & h_e(M-3) & \dots & h_e(0) \end{pmatrix} \rightarrow (22)$$

In this equation, the rows are related by a circular shift to the right. i.e., the i^{th} row elements can be obtained from the $i-1^{\text{th}}$ row by shifting its elements circularly right by one position.

A square matrix in which each row is a circular shift of the preceding row and the first row is a circular shift of the last row. It is called as “circulant matrix”.

For example,

Assume $A = 3$ and $B = 2$

$M \geq A + B - 1$

$M \geq 3 + 2 - 1 = 4$ and then append one zero to the samples of $f(x)$ and two zeros to samples of $h(x)$

In this, f and g are 4D vectors and H is 4×4 matrix.

$$H = \begin{pmatrix} h_e(0) & h_e(3) & h_e(2) & h_e(1) \\ h_e(1) & h_e(0) & h_e(3) & h_e(2) \\ h_e(2) & h_e(1) & h_e(0) & h_e(3) \\ h_e(3) & h_e(2) & h_e(1) & h_e(0) \end{pmatrix}$$

However, as $h_e(x) = 0$ for $x=2, 3$ and $h_e(x) = h(x)$ for $x=0, 1$

$$H = \begin{pmatrix} h_e(0) & & & h_e(1) \\ h_e(1) & h_e(0) & & \\ & h_e(1) & h_e(0) & \\ & h_e(1) & h_e(0) & \end{pmatrix}$$

where, all the elements not indicated in the matrix are zero.

The extension of the concepts for 2-D discrete degradation model is straight forward.

For two digitized images $f(x, y)$ and $h(x, y)$ of sizes $A \times B$ and $C \times D$ respectively, extended images of size $M \times N$ may be formed by padding with zeros and they may be given by

$$f_e(x, y) = \begin{cases} f(x, y) & 0 \leq x \leq A-1 \text{ and } 0 \leq y \leq B-1 \\ 0 & A \leq x \leq M-1 \text{ and } B \leq y \leq N-1 \end{cases} \rightarrow (23)$$

And

$$h_e(x, y) = \begin{cases} h(x, y) & 0 \leq x \leq C-1 \text{ and } 0 \leq y \leq D-1 \\ 0 & C \leq x \leq M-1 \text{ and } D \leq y \leq N-1 \end{cases} \rightarrow (24)$$

for $x = 0$ to $M-1$

$y = 0$ to $N-1$

The complete degradation model is obtained by adding the noise term $n_e(x, y)$. It is given as

$$g(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f_e(m-n) h_e(x-m, y-n) + n_e(x, y)$$

for $x = 0, 1, 2, 3 \dots M-1$ &

$y = 0, 1, 2, 3 \dots M-1$

Let f , g & n represent MN -dimensional column vectors formed by the rows of $M \times N$ functions $h_e(x, y)$, $g_e(x, y)$ and $n_e(x, y)$.

The elements in the first row of $f_e(x, y)$, the next N elements are from the second row and so on for all M rows of $f_e(x, y)$.

This allows the equation, $g = Hf + n$ $\rightarrow (25)$

Where f , g and n are of dimension of $MN \times 1$ and H is of dimension $MN \times MN$. This matrix consists of M^2 partitions, each partition being of size $M \times N$ and ordered according to

$$H = \begin{pmatrix} H_0 & H_{M-2} & H_{M-2} & \dots & H_1 \\ H_1 & H_0 & H_{M-1} & \dots & H_2 \\ H_2 & H_1 & H_0 & \dots & H_3 \\ \dots & \dots & \dots & \dots & \dots \\ H_{M-1} & H_{M-2} & H_{M-3} & \dots & H_0 \end{pmatrix} \rightarrow (26)$$

Each partition h_j is constructed from the j^{th} row of the extended function $h_e(x, y)$

As follows,

$$H = \begin{pmatrix} h_e(j, 0) & h_e(j, N-1) & \dots & h_e(j, 1) \\ h_e(j, 1) & h_e(j, 0) & \dots & h_e(j, 2) \\ \dots & \dots & \dots & \dots \\ h_e(j, N-1) & h_e(j, N-2) & \dots & h_e(j, 0) \end{pmatrix} \rightarrow (27)$$

IMAGE NOISE MODELS

Error or uncertainty of an image arising from such sources as sensor noise, film grain irregularities, and atmospheric light fluctuations. These all such effects are called as “noise”.

Noise Modelling

During image transmission, the noise can be occurs in the digital images. The performance of image is affected by various factors such as environmental conditions and quality of the sensing elements themselves.

For example, an image transmitted through a wireless network may be corrupted as a result of lightning or other atmospheric disturbances.

Spatial and Frequency Properties of Noise:

Frequency properties refer to the frequency content of noise in the Fourier sense (i.e. as opposed to frequencies of electromagnetic spectrum). For example, when the Fourier Spectrum of noise is constant, the noise is called “white noise”.

Spatial properties refer to the noise is independent of spatial coordinates and it is uncorrelated with respect to the image itself (i.e. there is no correlation between pixel values and the values of noise components)

Some important Noise Probability Density Function:

i) Gaussian Noise:

The PDF of a Gaussian random variable 'z', is given by

$$P(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z - \bar{z})^2}{2\sigma^2}}$$

where, Z=Intensity

$\bar{z}$ = Mean (average) value of Z

σ = Its standard deviation

σ^2 = Variance of Z

ii) Rayleigh Noise:

The PDF of a Rayleigh noise is given by

$$P(z) = \begin{cases} \frac{2}{b} (z - a) e^{-(z-a)^2/b} & \text{for } z \geq a \\ 0 & \text{for } z < a \end{cases}$$

The mean and variance of this density are given by

$$\bar{z} = a + \sqrt{\frac{\pi b}{4}}$$

$$\text{And } \sigma^2 = \frac{b(4 - \pi)}{4}$$

iii) Erlang (Gamma) Noise:

The PDF of Erlang noise is given by

$$P(z) = \begin{cases} a^b z^{b-1} / (b-1)! e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where, the parameters are such that $a > 0$, b is a positive integer. The mean and variance are given by

$$\bar{z} = \frac{b}{a}$$

$$\text{and } \sigma^2 = b/a^2$$

iv) Exponential Noise:

The PDF of exponential noise is given by

$$P(z) = \begin{cases} a e^{-az} & \text{for } z \geq 0 \\ 0 & \text{for } z < 0 \end{cases}$$

where, $a > 0$

The mean and variance are

$$\bar{z} = \frac{1}{a}$$

$$\text{and } \sigma^2 = 1/a^2$$

v) Uniform Noise:

The PDF of uniform noise is given by

$$P(z) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq z \leq b \\ 0 & \text{otherwise} \end{cases}$$

The mean and variance are

$$\bar{z} = \frac{1}{b-a}$$

$$\text{and } \sigma^2 = (b-a)^2/12$$

vi) Impulse (salt and pepper) Noise:

The PDF of impulse noise is given by

$$P(z) = \begin{cases} p_a & \text{for } z = a \\ p_b & \text{for } z = b \\ 0 & \text{otherwise} \end{cases}$$

If $b > a$, intensity 'b' will appear as a light dot in the image otherwise 'a' will appear like a dark dot. If either p_a or p_b is zero, the impulse noise is called 'unipolar'. If neither probability is zero and if they are approximately equal, impulse noise values will resemble salt and pepper granules randomly distributed over the image.

For this reason, bipolar noise is called salt and pepper noise.

Periodic Noise:

Periodic noise in an image arises from electrical or electromechanical interference during image acquisition. This noise is the spatially dependent noise. Periodic noise can be reduced significantly through frequency domain filtering. For example if the image is severely corrupted by (spatial) sinusoidal noise of various frequencies. The Fourier transform of a pure sinusoidal noise of various frequencies. The Fourier transform of a pure sinusoid is a pair of conjugate impulses located at the conjugate frequencies of the sine wave.

$$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow \frac{1}{2j} [\delta(u + Mu_0, V + NV_0) - \delta(u - Mu_0, V - NV_0)]$$

Thus if the amplitude of a sine wave in the spatial domain is strong, we would expect the spectrum of the image is a pair of impulses for each sine wave in the image.

Photo-Detector Noise

Types of noises:

1. Photo-detector noise
2. Film grain noise

We can see these noises one by one.

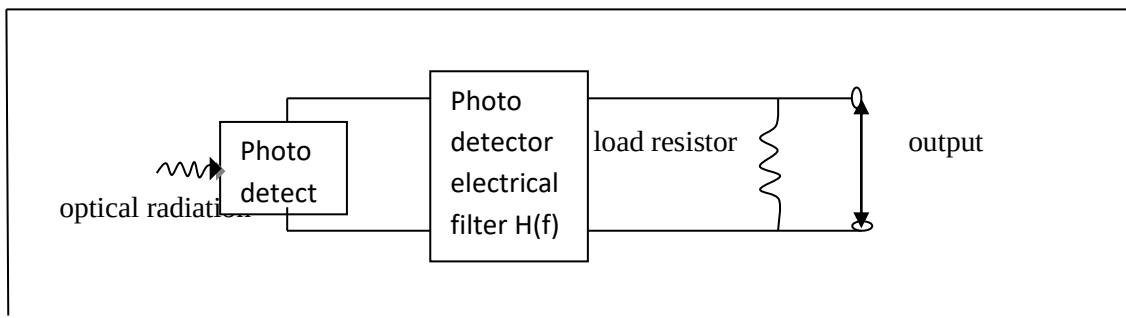


Fig.3.2.Photodetector

Fig.3.2. shows the photo-detector that contains photo-detector imaging sensor and an associated electrical filter that will serve as a basis for the discussion of sensor noise sources.

Ideally, the photo-detector acts as a converter of optical intensity to detector current or equivalently, as a converter of incident photons to electrons emitted by the detector.

The detector produces a signal current that passes through an electrical filter and creates a signal voltage across a load.

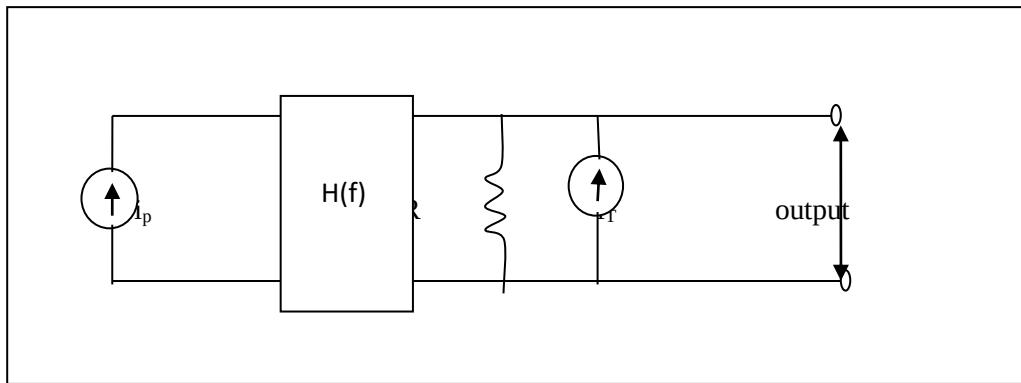


Fig.3.3.Photo-detector signal and thermal noise model

Thermal noise is the most common noise source associated with electronic imaging systems, resulting from random electron fluctuation in resistive elements of photo-detectors or resistors with sensor amplifiers.

Thermal noise can be modeled as an additive Gaussian stochastic process independent of the sensed image field.

As an illustration of thermal noise, calculations consider the photo-detector model of fig.in which the load resistor is assumed to be the only significant resistive element in the unit.

In the figure the photo-detector is replaced by an ideal current generator whose current is linearly proportional to the incident light intensity at some image point.

The thermal noise contribute can be modeled as a thermal noise current source of value i_T in parallel with the noise resistor.

The thermal noise current represents a zero mean Gaussian random process with variance $\sigma_{i_T}^2 = N_T / R$

where N_T is the thermal noise power at the system output

$$N_T = 2kT \int_{-\infty}^{\infty} |H_E(f)|^2 df$$

Where k is the Boltzmann's constant

$$k = 1.38 \times 10^{-23} \text{ Joules / degree Kelvin}$$

T is the temperature in degrees Kelvin and $H_E(f)$ is the equivalent transfer function of the photo-detector electrical filter and load resistor.

By simply, the filter is a capacitor of value C placed in parallel with the detector and the load resistor, the thermal noise power becomes,

$$N_T = kT/RC$$

If the detector current is constant and the equivalent filter is linear, the probability density of the current output of the detector can be written as

$$p(i) = (2\pi\sigma_{iT}^2)^{-1/2} \exp[-(i-i_s)^2/2\sigma_{iT}^2]$$

Where i_s is the equivalent average signal current at the output produced by the photo-detector current i_p passing through the filter.

Photo-detector current is not truly constant valued even when the incident light intensity is constant. Photoelectric sensors produces uncertainty resulting from the quantum mechanical nature of light.

At low light levels, the number of electrons emitted by a photo-detector is governed by a poisson probability density. In an observation time period " T ", the probability distribution of detector current pulses is given by

$$P_R [i_p = j_q / T] = (u_s + u_H)^j \exp\{-(u_s + u_H)\} / j!$$

Where j is a positive integer

q is the charge of an electron

$$q = 1.6 \times 10^{-19} \text{ coulombs}$$

u_s is the average number of electrons emitted front the detector as a result of the incident illumination and

u_H is the average number of electrons emission caused by detector dark current and extraneous background radiation.

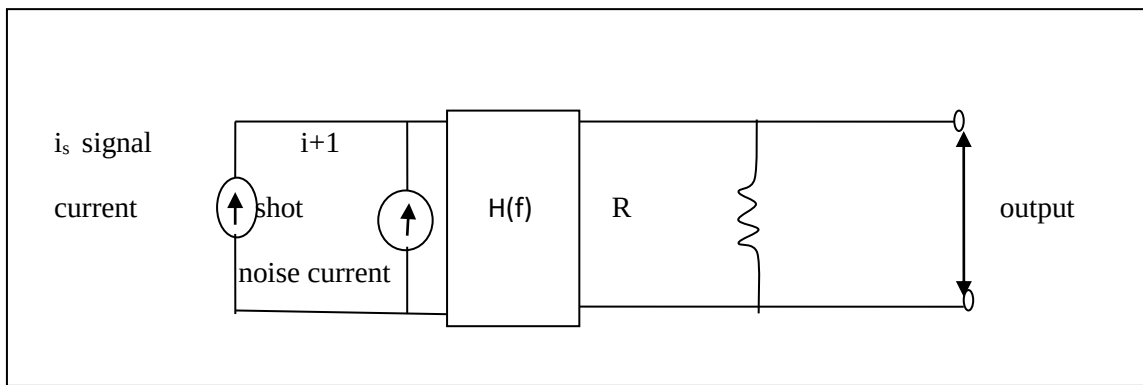


Photo-detector signal and short noise model

The resultant uncertainty in knowledge of the detector current is called shot noise in photo-emissive detectors and generation –recombination noise in photo-conductive and photovoltaic detectors.

If the photo-detector possesses sufficient internal amplification, for example, a photo-multiplier tube, the shot noise will normally dominate any subsequent thermal noise sources otherwise considered constant.

In either case, if the average number of signal electron emissions become large, the poisson distribution can be approximated by the Gaussian distribution.

$p(i_p) = [2\pi q^2 u_s / T^2]^{-1/2} \exp\{-i_p - q u_s / T\} / 2q u_s / T$ For most imaging sensors, the associated sensor can be modeled either as a Gaussian distributed or Poisson distribution random process.

Poisson distribution noise only occurs when the image light level is extremely small and the photo-detector possesses a large internal electron amplification.

Film Grain Noise

Silver halide grains are used in the exposure and development of a photographic film, that are exposed to a sufficient quantity of light are converted to grains of metallic silver.

This process is not entirely deterministic, silver halide grains experiencing an equivalent exposure are not necessarily converted to the same size and shape silver grains, furthermore silver grains are randomly distributed over the surface of the film.

This inherent randomness in silver grain formation is called “film – grain noise”, leads to a randomness or uncertainty in the amount of light passing through a transparency or reflected from a print.

The classical model for film grain noise fluctuations assumes a Gaussian distribution of the exposed film density $D(x, y)$

$$P[D(x, y)] = [2\pi\sigma_D^2(x, y)]^{-1/2} \exp\left\{-\frac{[D(x, y) - u_D(x, y)]^2}{2\sigma_D^2(x, y)}\right\}$$

About a mean density $u_D(x, y)$ with a variance $\sigma_D^2(x, y)$

The mean density is determined by a spatial average about the point (x, y) over some window including many individual grains and the standard deviation is modeled as

$$\sigma_D(x, y) = \alpha [u_D(x, y)]^\beta$$

where β is a constant

α is empirically set at

$$\alpha = 0.66 \left(\frac{a}{A} \right)^{1/2}$$

where 'a' is the average film grain area and 'A' is the examination area of the film Falconer has suggested $\beta = 1/2$ while Huang has setting $\beta = 1/3$.

The density $D(x, y)$ described by equation can be expressed as the sum of a mean density components $u_D(x, y)$ and a zero mean Gaussian random process component $N(x, y)$ according to the relation.

$$D(x, y) = u_D(x, y) + \sigma_D(x, y) N(x, y)$$

or
$$D(x, y) = u_D(x, y) + \alpha [\sigma_D(x, y)]^\beta N(x, y)$$

The film grain noise is signal dependent, it is additive in the density domain. The transmittance of the exposed film at a fixed wavelength may be written as

$$\tau(x, y) = 10^{-D(x, y) D(\lambda)}$$

Where $D(\lambda)$ is the characteristic density of the film as a function of wavelength.

Then it is easily seen that the transmittance function may be expressed as the product of the average transmittance without grain noise $\tau_0(x, y)$ and a noise factor proportional to $\sigma_D(x, y)$.

$$\tau(x, y) = \tau_0(x, y) 10^{-\sigma D(x, y) N(x, y) D(\lambda)}$$

Thus in the intensity domain, film grain noise is a multiplicative noise process.

SPATIAL FILTERING

The use of a spatial mark for image processing is called spatial filtering.

Two types of spatial filtering

- i) Linear filters,
- ii) Non Linear filters.

i) Linear Spatial Filter:

Linear filter means that the transfer function and the impulse or point spread function of a linear system are inverse Fourier transforms of each other.

Three types of linear filters are,

- a) Lowpass filters,
- b) Highpass filters, and
- c) Bandpass filter.

a)Lowpass filter:

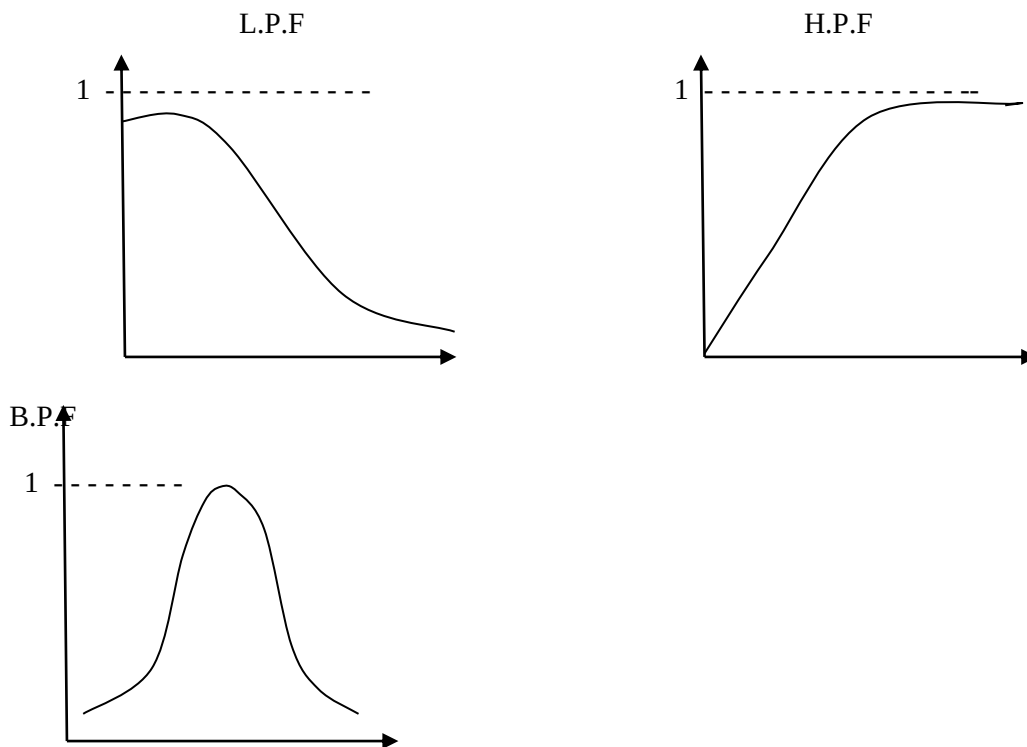
It attenuate or eliminate high frequency components in the Fourier domain. High frequency components characterize edges and others sharp details in an images, so the low pass filter causes image blurring.

b)Highpass filters:

It attenuate or eliminate low frequency components. These components are responsible for slowly varying characteristics of an image, such as overall contrast and average intensity.

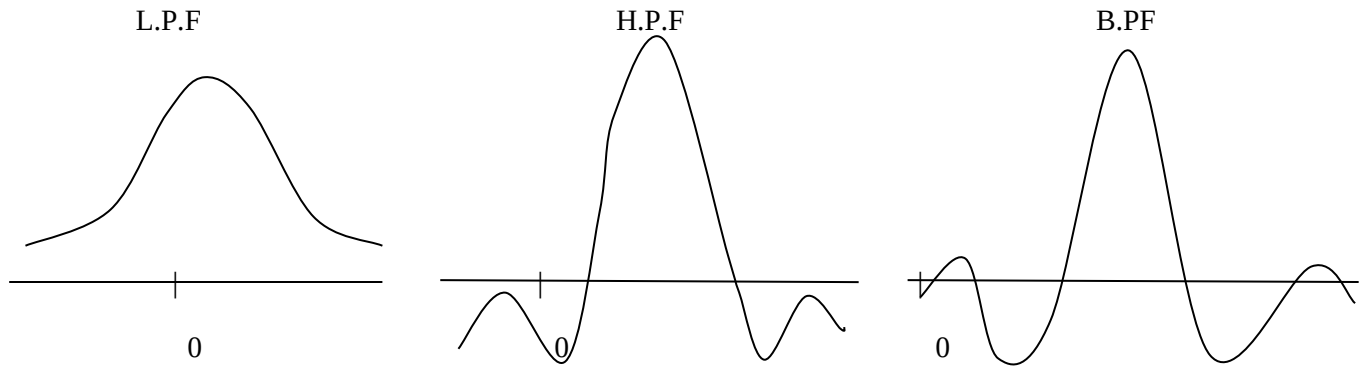
c)Bandpass filter:

It removes selected frequency regions between low and high frequencies. It is used for image restoration.



(A) Frequency Domain Filters

Fig.(a) represents the cross sections of circularly symmetric lowpass, highpass and bandpass filters in the frequency domain and their corresponding spatial filters are shown below in **fig(b)**.



(a) Spatial Domain Filters

The linear mask is the sum of the products between the mask coefficients and the intensities of the pixels under the mask at a specific location in the image.

It is denoted by the gray levels of pixels under the mask at any location by $z_1, z_2, \dots, z_9$. The response of a linear mask is

$$R = w_1 z_1 + w_2 z_2 + \dots + w_9 z_9.$$

3x3 mask coefficient is

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

ii) Nonlinear spatial filter:

This filter is based directly on the values of the pixels in the neighborhood and they did not use coefficients.

Using nonlinear filter noise reduction can be achieved effectively.

The basic function of the nonlinear filter is to compute the median gray level value in the neighborhood in which the filter is located.

Other examples of nonlinear filters are max and min filter.

- Max filter is used to find the brightest points in an image.
- Min filter is the opposite of max filter.

SMOOTHING FILTERS:

Smoothing filters are used for blurring and noise reduction.

Blurring is used in preprocessing steps, such as removal of small details from large image and bridging of small gaps in lines and curves.

Noise reduction can be achieved by blurring with a linear filter and by nonlinear filter.

Types of Smoothing filters:

Lowpass Spatial Filtering

It shows that, the LP spatial filter have all positive coefficients.

From this equation,

$$R = w_1z_1, w_2z_1, \dots, w_9z_9.$$

The response will be the sum of gray levels of nine pixels which could cause R to be out of the valid gray level range.

Therefore the solution is to scale the sum by dividing R by 9.

Generally, the response R would simply be the average of all the pixels in the area of mask. Spatial lowpass filters of various sizes are shown in fig.

1/9 x

1	1	1
1	1	1
1	1	1

(a)

1/25 x

1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1

(b)

It is referred to as neighborhood averaging.

MEDIAN FILTERING

If $Q=0$, this filter is reduced to the arithmetic mean filter. If $Q=-1$, this filter is reduced to harmonic filter.

In this filter, the value of the pixel is replaced by the intensity level of the neighborhood of that pixel.

The restored function is given as,

$$\hat{f}(x,y) = \text{Median}[g(u,v)]$$
$$(u,v) \leftarrow S_{xy}$$

Median filters are used to reduce certain types of random noise excellently. These are effective in the presence of unipolar and bipolar impulse noises.

The median operator is a nonlinear filter. It has the following properties.

- i) The median filter reduces the variance of the intensities in the image.
- ii) Median filters can change the image intensity mean value, if the spatial noise distribution in the image is not symmetrical within the window.
- iii) Median filters preserve certain edge shapes.
- iv) Given a symmetrical window shape the median filter preserves the location of edges.
- v) After applying a median filter, new gray values are not generated.

Computation of median filter requires sorting of a list of numbers.

For large window sizes, computation time is large.

One approach for neighbouring window locations is based on modification of the sorted list at the previous location (x,y) .

As we move the “ $n \times m$ ” window, we discard “ n ” points and add ‘ n ’ new points and leave the remaining “ $mn-2n$ ” points unchanged.

This approach is efficient only, if the amount of common image intensity data is significantly large from one position to the next.

The second approach which is based on list sorting is to calculate the median iteratively, by bit position, starting with MSB.

After determining the median bit for a bit position. It is identified that, only intensity values with this bit are retained for subsequent calculations.

This procedure terminates for ‘ n ’ bit data after n iterations, regardless of the size of the window.

Steps:

- i) Input the values in decimal numbers.
- ii) Convert them into binary form.
- iii) Find the median value of all the MSBs.
- iv) Find the median value of the next LSB.
- v) Continue the procedure of finding the median value of the next LSB bit, till all the bits are exhausted.
- vi) Write the median bits, starting from that of MSB to LSB. This is the overall median value.

DIRECTIONAL SMOOTHING

A directional averaging filter is used to protect the edges from blurring while smoothing. Spatial averages $V(m, n; \theta)$ are calculated in various directions. Its equation is given below.

$$V(m, n; \theta) = 1/N \sum_{(k,l) \in w_\theta} y(m-k, n-l)$$

The direction θ^* at which $|y(m, n) - V(m, n; \theta^*)|$ is minimum is noted. $V(m, n) = V(m, n; \theta^*)$ can give the desired result.

Sharpening Filters

Sharpening is used to highlight fine detail in an image.

Highpass Spatial Filtering

This highpass spatial filter indicates that, the impulse response of the filter should have positive coefficients near its center and negative coefficients in the outer periphery. For example consider, 3x3 mask

1/9 x	-1	-1	-1
	-1	8	-1
	-1	-1	-1

A highpass Spatial Filter

Here, sum of coefficients is '0'. Thus, when the mask is over an area of constant or slowly varying gray level, the output of the mask is zero or very small.

Reducing the average value of an image to zero implies that the image must have some negative gray levels.

If we use positive gray levels, the result of highpass filtering involve some form of scaling or clipping, so the gray levels range of [0, L-1] is used.

Therefore, positive values are not good. Negative value is better.

High- Boost Filtering

To obtain the better result, rather than highpass spatial filtering, we use high-boost filtering.

The highpass filter image is

$$\text{Highpass} = \text{Original image} - \text{Lowpass filter image}.$$

The definition of a highboost or high-frequency emphasis filter is given by multiplying the original image by an amplification factor denoted by 'A'.

$$\begin{aligned}\text{Highboost} &= (A)(\text{Original image}) - \text{Lowpass filter image} \\ &= (A-1)(\text{Original image}) + \text{original image} - \text{Lowpass filter image.} \\ &= (A-1)(\text{original}) + (\text{Highpass filter image}).\end{aligned}$$

When $A=1$, It gives the standard highpass result.

When $A>1$, It gives the highboost image which looks more like the original image.

The general process of subtracting a blurred image from an original is given below.

$\text{High} = (A)(\text{original}) - \text{Lowpass}$
This is known as “Unsharp masking”.

The center weight of the mask is

$$w = A - 1$$

$A \geq 1$ which determines the nature of the filter.

$$\frac{1}{9} \quad \times \quad \begin{array}{|c|c|c|} \hline -1 & -1 & -1 \\ \hline -1 & w & -1 \\ \hline -1 & -1 & -1 \\ \hline \end{array}$$

Mask for highboost spatial filter

Derivative Filters

Averaging of pixels produces blurred image.

Averaging is analogous to integration. But differentiation produces the opposite effect. i.e. Sharpen the image.

The most common method of differentiation in image processing application is the ‘gradient’.

For a function $f(x,y)$, the gradient of f at coordinates (x,y) is defined as the vector

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}$$

The magnitude of this vector is

$$|\nabla f| = \text{mag}(\nabla f) = [(\frac{\partial f}{\partial x})^2 + (\frac{\partial f}{\partial y})^2]^{1/2}$$

Consider the image region

Z_1	Z_2	Z_3
Z_4	Z_5	Z_6

Z ₇	Z ₈	Z ₉

Where, Z's denote the values of gray level equation above can be approximated at point Z₅ in a number of ways.

The simplest approach is to use the difference (Z₅ - Z₈) in the x direction and (Z₅ - Z₆) in the y direction. These are combined as,

$$\nabla f \approx [(Z_5 - Z_8)^2 + (Z_5 - Z_6)^2]^{1/2}$$

Take magnitude

$$\nabla f \approx |Z_5 - Z_8| + |Z_5 - Z_6|$$

Another method is to use cross differences in equation

$$\nabla f \approx [(Z_5 - Z_9)^2 + (Z_6 - Z_8)^2]$$

$$\nabla f \approx |Z_5 - Z_9| + |Z_6 - Z_8|$$

The above equations can be implemented by using the masks of size 2x2.

For example, these equation can be implemented by taking the absolute value of the response of the two masks shown below and summing the results. These masks are called “**Roberts cross-gradient operators**”.

1	0
0	-1

0	1
-1	0

If mask is 3x3, the neighbourhood is

$$\nabla f \approx |(Z_7 + Z_8 + Z_9) - (Z_1 + Z_2 + Z_3)| + |(Z_3 + Z_6 + Z_9) - (Z_1 + Z_4 + Z_7)|$$

The difference between the 3rd and 1st row of the 3x3 region approximates the derivative in the x direction and difference between the 3rd and 1st column approximates the derivative in the y direction called “prewitt operators” shown in the figure below.

-1	-1	-1
0	0	0
1	1	1

-1	0	1
-1	0	1
-1	0	1

Prewitt operators

For approximating the magnitude of the gradient, another pair of masks can be used which are known as sobel operators.

-1	-2	-1
0	0	0
1	2	1

-1	0	1
-2	0	2
-1	0	1

Sobel operators

Laplacian Filters

Laplacian filters is used to enhance discontinuities.

The laplacian of two dimensional image $f(x,y)$ is defined as,

$$\nabla^2 f(x,y) = [\partial^2 f / \partial x^2 + \partial^2 f / \partial y^2]$$

For a 3x3 sub image , the digital form equivalent to the Laplacian operator is given as

$$\nabla^2 f(x,y) = 4P_5 - (P_2 + P_4 + P_6 + P_8)$$

From this equation , it is possible to define the digital laplacian mask. So , the coefficient associated with center pixels should be positive and that associated with the other pixels should be negative.

Moreover, the sum of the coefficients should be zero.

The Laplacian response is sensitive to noise and is rarely used for edge detection.

The 3x3 kernel is

(a)

P ₁	P ₂	P ₃
P ₄	P ₅	P ₆
P ₇	P ₈	P ₉

(b)

0	-1	0
-1	4	-1
0	-1	0

The mask is used to compute Laplacian.

Frequency Domain Filters

Enhancement in frequency domain is achieved by computing the Fourier transform of the image to be enhanced. Then the result is multiplied by a filter transfer function and the inverse transform to produce the enhanced image.

There are three types of frequency domain filters.

- i) Smoothing filters
- ii) Sharpening filters
- iii) Homomorphic filters

Smoothing filters

Lowpass filters are used for noise smoothing and interpolation.

Highpass filters are used for extracting edges and in sharpening images.

Bandpass filters are used for enhancement of edges and other high pass characteristics in the presence of noise.

Lowpass filtering

Smoothing effect is achieved in the frequency domain by attenuating a specified range of high frequency components in the transform of a given image.

The low pass filter can be represented as $G(u,v) = H(u,v) F(u,v)$

Where,

$F(u,v)$ is the Fourier transform of an image to be smoothed.

$H(u,v)$ is the filter transfer function and

$G(u,v)$ is the enhanced image in the frequency domain.

In order to get the enhanced image in the spatial domain, the inverse transform is applied and is given by,

$g(x,y) = \text{Inverse Fourier transform of } [H(u,v) F(u,v)]$ In the equation $G(u,v)$ we have considered the transform function $H(u,v)$ that gives $Z(u,v)$ by attenuating the high frequency components of $F(u,v)$.

In general, most of the filter transfer functions affect the real and imaginary parts of $F(u,v)$ in the same manner.

These filters are called as “**Zero phase shift filters**”. Because, they do not change the phase of the transform.

Ideal low pass filter

The two dimensional ideal pass filter transfer function can be given by

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

Where, D_0 is a specified non negative quantity and

$D(u,v)$ is the distance from point (u,v) to the origin of the frequency plane

i.e. $D(u,v) = \sqrt{u^2 + v^2}$

(a) Perspective plot of an ideal L.P.F transfer func.

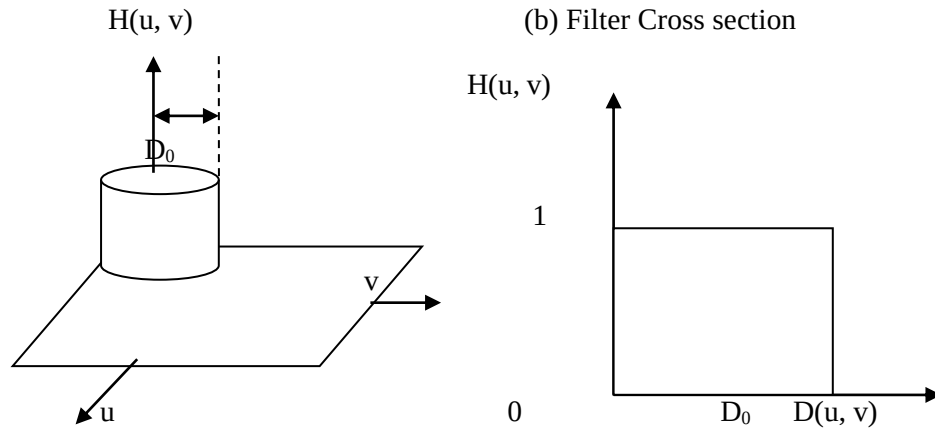


Figure shows that, the filter passes all the frequencies inside the circle of radius D_0 where it attenuates all the frequencies outside this circle. Hence, this filter is called “Ideal low pass filter”.

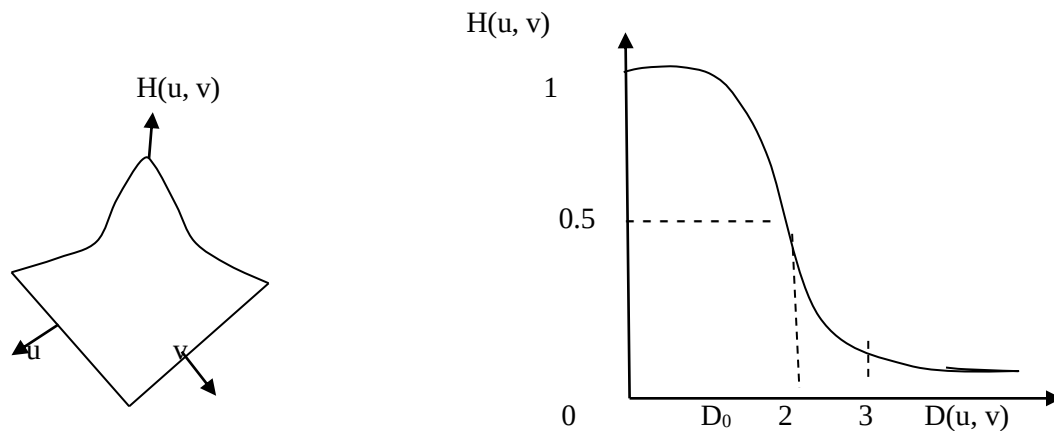
Butterworth lowpass filter

The response of the butterworth low pass filter of order n is defined by the equation

$$H(u, v) = 1 / [1 + (D(u, v) / D_0)^{2n}]$$

Where $D(u, v)$ is the distance from the point (u, v) to the origin and is given by $\sqrt{u^2 + v^2}$

The 3-D and cross sectional views of the butterworth low pass filter response are shown below.



When $D(u, v) = D_0$, $H(u, v) = 0.5$. This indicates that, at cutoff frequency the response is half of its maximum value.

In the most of the cases, at cutoff frequency, the response will be equal to $1/\sqrt{2}$ times the maximum value of $H(u, v)$. To have this effect, the equation should be modified as

$$H(u, v) = 1 / [1 + (\sqrt{2} - 1) (D(u, v) / D_0)^{2n}]$$

$$H(u,v) = 1 / (1 + 0.414 [D(u,v) / D_0]^{2n})$$

SHARPENING FILTER

The highpass filter is useful in extracting edges and in sharpening images.

The highpass filter passes the high frequency components and it attenuates low frequency components corresponding to slow varying details of the image.

The ideal highpass filter which has sharp or abrupt transition is given by the equation

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \leq D_0 \\ 1 & \text{if } D(u,v) > D_0 \end{cases}$$

Where, D_0 is the cutoff frequency which is measured from the origin and

$D(u,v)$ is the distance from the origin and is given by

$$D(u,v) = \sqrt{u^2 + v^2}$$

The ideal low pass filter which has the abrupt transition at the cutoff frequency cannot be realized using the electronic circuit components. However, it can be realized with smooth transition frequency and such filters are called as Butterworth filters.

The transfer function of the high pass butterworth filter of order n with a cutoff frequency D_0 from the origin is given by

$$H(u, v) = \frac{1}{1 + [D_0 / D(u,v)]^{2n}}$$

HOMOMORPHIC FILTERS:

We know that, the 'illumination' and 'reflectance' components of the image light intensity function $f(x, y)$ are denoted by $i(x, y)$ and $r(x, y)$ respectively.

In frequency domain, it is useful for improving the appearance of an image by using simultaneous brightness range compression and contrast enhancement.

$f(x, y)$ is given as,

$$f(x, y) = i(x, y) \cdot r(x, y) \quad \rightarrow (1)$$

where, $0 < i(x, y) < \infty$ and

$$0 < r(x, y) < 1$$

The equation cannot be used directly, in order to operate separately on the frequency components of illumination and reflectance. Because, in frequency domain, the Fourier transform of the product of two functions is not separable.

$$\text{i.e. } F\{f(x, y)\} \neq F\{i(x, y)\} \cdot F\{r(x, y)\}$$

but we can define

$$\begin{aligned} z(x, y) &= \ln f(x, y) \\ &= \ln i(x, y) + \ln r(x, y) \end{aligned}$$

Then, we can write

$$\begin{aligned} F\{z(x, y)\} &= F\{\ln f(x, y)\} \\ &= F\{\ln i(x, y)\} + F\{\ln r(x, y)\} \end{aligned}$$

Otherwise

$$Z(u, v) = I(u, v) + R(u, v)$$

Where $I(u, v)$ and $R(u, v)$ are the Fourier transforms of $\ln i(x, y)$ and $\ln r(x, y)$ respectively.

We already know that, in frequency domain the convolution of an image $f(x, y)$ and linear position invariant operator $h(x, y)$ is given in Fourier transform as

$$G(u, v) = H(u, v) \cdot F(u, v)$$

$$H(u, v) = \text{Homomorphic filter function}$$

Write $z(u, v)$ by means of a filter function $H(u, v)$ from above equation.

$$S(u, v) = H(u, v) S(u, v)$$

$$S(u, v) = H(u, v) I(u, v) + H(u, v) R(u, v)$$

Where $S(u, v)$ is the Fourier transform of the result.

In spatial domain representation

$$\begin{aligned} S(x, y) &= F^{-1}\{S(u, v)\} \\ &= F^{-1}\{H(u, v) I(u, v)\} + F^{-1}\{H(u, v) R(u, v)\} \rightarrow (2) \end{aligned}$$

$$\text{Let } i'(x, y) = F^{-1}\{H(u, v) I(u, v)\}$$

$$\text{And } r'(x, y) = F^{-1}\{H(u, v) R(u, v)\}$$

Then equation 2 becomes

$$s(x, y) = i'(x, y) + r'(x, y)$$

$Z(x, y)$ is formed by taking the logarithm of the original image $f(x, y)$. The inverse operation yields the desired enhancement image $g(x, y)$.

So, take the exponential function

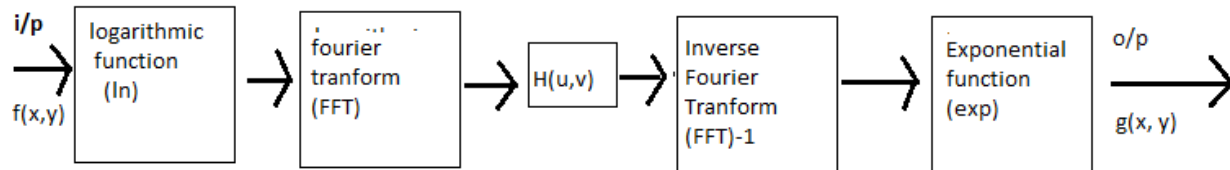
$$\begin{aligned}\therefore g(x, y) &= \exp[s(x, y)] \\ &= \exp[i'(x, y)] \cdot \exp[r'(x, y)]\end{aligned}$$

$$g(x, y) = i_o(x, y) \cdot r_o(x, y)$$

where, $i_o(x, y) = \exp[i'(x, y)]$ and

$$r_o(x, y) = \exp[r'(x, y)]$$

are the illumination and reflectance components of the output image.



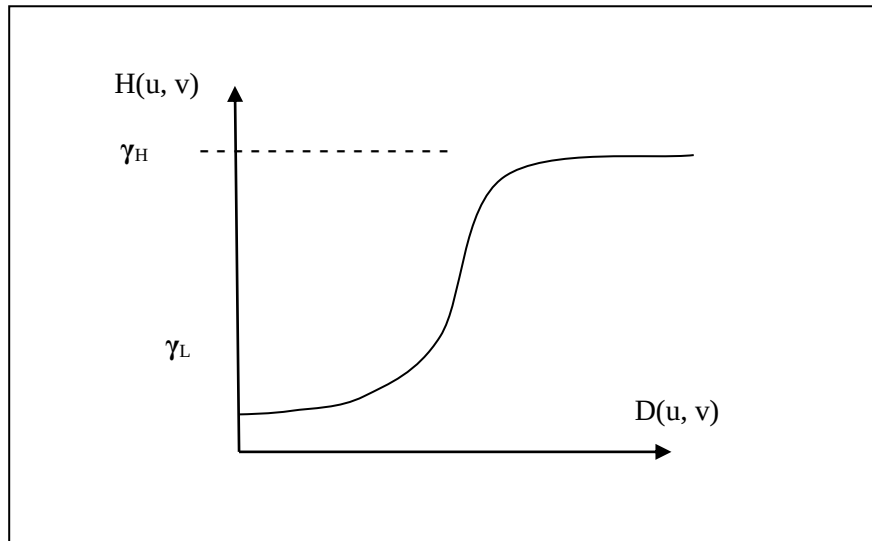
Homomorphic filtering approach for image enhancement

The diagram shows the above said enhancement techniques. This method is called as 'homomorphic systems'.

In this system, the main thing is that separation of illumination and reflectance components. Then, homomorphic filter function $H(u, v)$ can operate on these components separately.

The filter function $H(u, v)$ that affects the low frequency components of the Fourier transform with illumination and the high frequencies with reflectance in different ways.

So $H(u, v)$ is rotated by 360° cross section about vertical axis.



Cross section of circulatory symmetric filter function

$D(u, v) \rightarrow$ distance from the origin

The parameters γ_H and γ_L should be

$$\gamma_L < 1 \text{ and } \gamma_H > 1$$

The filter function shown in fig. is used to decrease the low frequencies and amplify the high frequencies. Finally, are simultaneously done dynamic range compression and contrast enhancement.